Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahaj-bhadja sahaj-bhadja

Summary

Software engineer with 2+ years building APIs, cloud workflows, and developer-facing platform features. Strong in Python, JavaScript/TypeScript, REST services, SQL-backed systems, CI/CD, and reliable customer-facing control-plane style functionality.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant platform services on Azure SQL and Blob Storage for 500+ users, optimizing indexed queries, access patterns, and configuration workflows to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant asynchronous workflows with FastAPI background workers, supporting 1K+ AI jobs through explicit job-state transitions, failure isolation, and dependable execution across concurrent production control platform flows.
- Designed integration pipelines across FastAPI, SQL Server, PostgreSQL, and cloud storage, processing 5K+ monthly jobs with validation checks, reconciliation logic, and CI/CD-backed releases for dependable platform operations.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask APIs for an AI grading product handling 10,000+ daily requests, integrating uploads, grading history, and secure file-processing workflows across fast-moving customer-facing platform requirements.
- Integrated multimodal services, concurrent batch processing, and production backend workflows on AWS, reducing feedback turnaround by 60% while maintaining stable API behavior and dependable service execution under peak usage.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform data-sharing software with a microservice architecture, supporting scalable service communication, distributed validation workflows, and maintainable backend components across Windows, Linux, and macOS shared production environments.
- Optimized API and ETL integrations using Python, SQL, AWS Glue, and S3, improving response time by 30% while strengthening pipeline efficiency, systems coordination, and operational behavior across production workloads.
- Maintained 20+ automated tests and CI pipelines with Jenkins, PyTest, and Mockito, increasing release reliability by 68% while improving debugging, deployment consistency, and defect tracking across distributed software builds.
- Executed multi-platform build and validation workflows across Windows, Linux, and macOS, helping surface defects earlier, verify service behavior consistently, and support repeatable releases for customer-facing platform components.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built MERN platform features with REST APIs, WebSockets, and responsive interfaces, supporting synchronized records across modular workflows while translating requirements into reliable end-to-end production behavior for customer-facing applications.

Skills

Languages: Python, JavaScript, TypeScript, SQL, Java, C++, Bash
APIs & Backend: FastAPI, Flask, Node.js, Express, REST APIs, WebSockets, background workers, microservices
Frontend: React, Next.js, HTML, CSS, TailwindCSS, responsive design, UI state, customer-facing workflows
Data & Cloud: PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, AWS, Azure, GCP, S3, AWS Glue, Blob Storage
Delivery: Git, GitHub, GitHub Actions, Jenkins, Docker, CI/CD, monitoring, debugging, operational support, code reviews
Engineering: object-oriented programming, data structures, algorithms, cloud concepts, configuration workflows, platform reliability

Projects

Selected Projects | *Python, AWS, Java, TypeScript, PostgreSQL*
- Developed and deployed a containerized RESTful data processing API on AWS with automated CI/CD releases, validation checks, and production-grade reliability, demonstrating scalable service delivery across cloud environments.
- Built an AI-powered SQL development workspace supporting 10+ databases, combining Java and TypeScript services to accelerate query authoring, schema exploration, and structured database management from a unified interface.
- Engineered a cloud-based ingestion and indexing platform processing 50K+ records, implementing incremental updates, metadata persistence, and performance tuning to reduce retrieval latency by 40% in production workflows.